

EECS 4080 Computer Science Project

Project Contract

Course: EECS 4080

Academic Year: Summer 2025

Project Title: Sentiment Analysis of Social Media Posts on Polarizing Topics

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Length: One term (Summer 2025)

Project Description

Social media plays a significant role in shaping public discourse around polarizing topics like politics, climate change, and AI ethics. These platforms often foster echo chambers that reinforce users' pre-existing beliefs.

This project aims to develop a real-time sentiment analysis system that scrapes posts from platforms like Twitter, Reddit, and YouTube, analyzes their sentiment using NLP models, and visualizes public opinion trends. The system will also incorporate predictive modeling to forecast future sentiment trajectories.

The project will involve building a web-based tool with the following components:

- **Data Collection:** Web scraping using tools like sncrape and BeautifulSoup.
- **Sentiment Analysis:** Using VADER, TextBlob, and BERT-based models for polarity and emotion detection.
- **Visualization:** Real-time interactive dashboards (React.js + Plotly/D3.js).
- **Prediction:** Implementing LSTM or ARIMA models for forecasting sentiment trends.

The goal is to provide a research-oriented system to track and interpret public sentiment on controversial issues, contributing insights for sociological research and digital media analysis.

Background Requirements:

Prerequisite Courses: EECS 4412 or equivalent knowledge in NLP or data analytics

Experience:

- Proficient in Python
- Experience with web scraping and NLP libraries (NLTK, TextBlob, HuggingFace Transformers)
- Basic frontend development knowledge (React.js)
- Familiarity with machine learning and visualization tools

Resources:

- **Software:** Python, Flask, React.js, NLTK, scikit-learn, Plotly, HuggingFace Transformers
- **Hardware:** Personal computer; access to cloud/GPU services if needed
- **Dataset:** Data scraped from Twitter, Reddit, YouTube, and news headlines

Readings:

- [1] Hutto, C.J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text.
- [2] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.
- [3] Pang, B., & Lee, L. (2008). Opinion Mining and Sentiment Analysis.
- [4] Bachate, M, and Suchitra, S. (2025). Sentiment analysis and emotion recognition in social media: A comprehensive survey, Applied Soft Computing, Vol. 174, April 2025, 112958 (<https://doi.org/10.1016/j.asoc.2025.112958>).
- [5] Paul, J., Das Chatterjee, A., Misra, D., Majumder, S., Rana, S., Gain, M., De, A., Mallick, S., Sil, J. (2024). A survey and comparative study on negative sentiment analysis in social media data, Multimedia tools and applications, Vol.83 (30), p.75243-75292.
- [6] Wankhade, M., Rao, A.C.S. & Kulkarni, C. (2022). A survey on sentiment analysis methods, applications, and challenges. Artificial Intelligence Review, 55, 5731–5780 (2022). <https://doi.org/10.1007/s10462-022-10144-1>.

Deliverables:

- A functioning web-based sentiment analysis tool
- Two interim reports
- A final report with system design, experiments, evaluation, and conclusions
- Documented source code and user guide

Milestone List (May-August 2025):

May 1 – May 15: Setup & Initial Scraping

- Set up GitHub repository and project environment

- Develop and test scraping scripts using snsrape (Twitter) and BeautifulSoup (Reddit)
- Begin collecting sample data based on predefined polarizing topics (e.g., climate change, AI ethics)

May 16 – May 31: Sentiment Analysis Engine

- Implement sentiment classifiers (VADER and TextBlob) to label scraped posts
- Preprocess and store labeled data for testing
- Experiment with integrating BERT for deep sentiment/emotion classification
- First progress report due on May 31 (Describing the progress so far)

June 1 – June 15: Backend API Development

- Build backend using Flask or FastAPI to expose endpoints for:
- Triggering scraping
- Running sentiment analysis
- Returning results as JSON
- Test API locally and prepare response formats for frontend integration

June 16 – June 30: Frontend Development (v1)

- Set up React.js frontend with a topic input field and submit button
- Integrate frontend with backend using Axios/fetch
- Display basic sentiment results (counts and charts) dynamically

July 1 – July 15: Visualization and Feature Expansion

- Add advanced components: sentiment-over-time graph, word cloud, and platform comparison

- Integrate charting libraries (e.g., Plotly, Chart.js, D3.js)
- (Optional) Incorporate emotion detection or topic modeling modules
- Second progress report due on May 31 (Describing the progress so far)

July 16 – July 31: Prediction & Hosting Preparation

- Implement forecasting feature using ARIMA or LSTM (if time permits)
- Finalize UI styling and interactivity
- Prepare dashboard for deployment (e.g., Vercel, Heroku, or Render)
- Conduct multi-browser testing and fix frontend/backend bugs

August 1 – August 15: Final Deployment & Polish

- Deploy final web dashboard with full functionality
- Finalize documentation (README.md, setup guide, user instructions)
- Ensure clean and modular code with adequate comments

August 16 – August 31: Final Presentation & Demonstration

- Prepare a live demo walkthrough of the tool
- Motivation, architecture, features, and results
- Submit deployed tool link, source code, and presentation
- Final report due on August 28

Evaluation:

Functionality & Technical Quality – 15%

Accurate implementation of web scraping, sentiment analysis models (VADER, TextBlob, BERT), backend integration, and overall system functionality. Includes successful handling of data pipelines and optional forecasting features.

User Interface & Visualization – 15%

Quality and clarity of the web dashboard interface. Includes effective use of visual elements such as pie charts, word clouds, sentiment trends over time, and comparison charts. Emphasis on interactivity, responsiveness, and user experience.

Deployment & Accessibility – 10%

Successful hosting of the dashboard on a public platform (e.g., Vercel, Heroku). Webpage should be accessible, browser-compatible, and include basic error handling for missing or delayed data.

Presentation & Demonstration – 30%

Delivery of a clear and engaging final demo. Explains project goals, system architecture, key features, and challenges. Includes a walkthrough of the live system and answers to supervisor questions.

Code Quality & Documentation – 10%

Readable and well-organized code with in-line comments. Includes a comprehensive README.md, setup guide, and usage instructions. GitHub repository should reflect consistent progress and version control.

Final Report – 20%

A report that describes what is done in the project, including introduction, methodologies taken, experimental results, discussions and conclusions.

Signatures:

Student:

Aditya Manjrekar

Supervisor:



Aijun An

Course director: _____